30 Azure Cloud Interview Questions Answer(Real, Company-Driven

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1️⃣ How do you design high availability in Azure for a global application?

Designing HA is not just “use multiple regions”. It is about eliminating every single point of failure across 5 layers:

🔷 Layered HA Model (Production Grade)

Layer	Design Strategy
DNS	Azure Front Door / Traffic Manager
Edge	CDN + WAF
App	Multi-region deployment
Compute	Zone redundant
Data	Geo-replicated

🔷 Real Production Scenario

You are building a global SaaS product (like Shopify) with users in:

- India

- Europe
- US

Architecture

- Azure Front Door
  - Health probes every 30 seconds
  - Geo routing (India → Central India, Europe → West Europe)
- App Layer
  - Deployed in 3 regions
  - Each region uses App Service with Zone Redundancy
- Database
  - Azure SQL → Active Geo Replication
  - Failover group configured
- Cache
  - Redis with geo-replication

🔷 Failure Simulation (Real thinking)

Case 1: VM crashes

- Azure auto-heals instance
- Load balancer removes unhealthy instance

Case 2: Entire zone fails

- Other zones serve traffic

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Case 3: Entire region fails (rare but real)

- Front Door routes traffic to nearest healthy region
- DB failover group switches read/write

🔷 Important Design Decisions

Decision	Reason
Stateless services	So any instance can serve request
External session store (Redis)	Avoid session loss
Health probes	Detect failure quickly
Async messaging	Avoid cascading failures

🔷 Common Mistakes Engineers Make

Single region with Availability Set
DB not geo-replicated
Hardcoded region endpoints
No chaos testing

2️⃣ When would you choose Azure App Service vs AKS — and why?

This is a very common senior-level decision.
The wrong decision can cost:

- Money
- Maintenance overhead
- Team productivity

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🔷 Decision Framework Used in Real Companies

Choose Azure App Service when:

- You are building:
  - Web apps
  - REST APIs
  - Backend for mobile apps
- You want:
  - Auto scaling
  - Built-in SSL
  - Zero infrastructure management

Example (Real)

You are building:
👉 Internal HR portal
👉 CRM system
👉 Admin dashboard
👉 Use App Service

Choose AKS when:

- You have:
  - 20+ microservices
  - Need sidecars (logging, auth)
  - Need service mesh (Istio)

Example (Real)

You are building:
👉 Netflix-like streaming backend
👉 Food delivery platform
👉 Ride sharing system
👉 Use AKS

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🔷 Cost Comparison Reality

Factor	App Service	AKS
Infra mgmt	None	High
Scaling	Simple	Complex
DevOps effort	Low	High
Flexibility	Limited	Full control

🔷 Real mistake seen in industry

A startup used AKS for 3 APIs
Result:

- High DevOps overhead
- Expensive cluster cost
- Slow delivery

They later migrated to App Service and reduced cost by 45%

3️⃣ Explain Azure Landing Zone and its importance in enterprises

Landing Zone is not optional in enterprise.
It is the foundation of everything in Azure.

🔷 What exactly is Landing Zone?

It is a pre-architected environment that includes:

- Identity
- Networking
- Security
- Governance
- Monitoring

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🔷 Enterprise Architecture (Real)

Large enterprise uses Hub-Spoke model

HUB

- Firewall
- VPN Gateway
- Shared services

SPOKES

- App environments
- Dev / QA / Prod

🔷 What Landing Zone enforces

- Only approved regions allowed
- Mandatory tagging
- Resource naming convention
- No public IPs allowed
- Secure networking

🔷 Real Scenario

Bank migrating to Azure:
Without Landing Zone:

- Dev teams created random VNets
- No firewall
- No logging

With Landing Zone:

- All VNets peered to central hub

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- All traffic goes through firewall
- Central monitoring enabled

🔷 Tools used

- Azure Policy
- RBAC
- Blueprints
- Defender for Cloud

4️⃣ How do you handle cost optimization in Azure at scale?

Cost optimization is continuous engineering activity, not one-time.

🔷 Real Production Cost Strategy

Step 1 — Visibility

- Azure Cost Management
- Tagging by:
    - Project
    - Environment
    - Owner

Step 2 — Identify Waste

Common waste:

- Unused disks

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- Idle VMs
- Over-provisioned DB

Step 3 — Optimization Techniques

1. Right-sizing

Example:

- DS5 VM used at 10% CPU → downgrade to DS2

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2. Reserved Instances

Save up to 70%
Used for:

- Databases
- App services

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3. Auto Scaling

Scale rules:

- CPU > 70% → scale out
- CPU < 30% → scale in

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4. Storage Tiering

- Hot → active data
- Cool → monthly access
- Archive → rarely used

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5. PaaS over IaaS

App Service cheaper than VM cluster

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🔷 Real Case Study

E-commerce company:
Before optimization:

- Monthly cost: ₹32 lakhs

After:

- Auto-scaling
- Reserved DB
- Removed unused disks

👉 Final cost: ₹19 lakhs

5️⃣ What happens internally when an Azure VM is restarted?

This is a fabric-level operation handled by Azure Compute Fabric.

🔷 Types of Restart

1. Soft Restart (Reboot)

- OS restart
- Same host
- Same IP
- No data movement

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2. Hard Restart (Stop + Start)

- VM deallocated
- Compute host released
- New host assigned

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🔷 Internal Flow (Deep level)

1. Azure Fabric Controller receives request
2. VM state marked as stopped/deallocated
3. Compute resources freed
4. When started:
    - New host allocated
    - Network attached
    - Disk reattached

🔷 Important Real Impacts

Impact	Result
Public IP (dynamic)	Changes
Ephemeral disk	Lost
Private IP	Usually same
Availability set	Maintained

🔷 Real Production Issue

A team used:

- Temporary disk for logs

After VM restart:
👉 All logs lost
Fix:
👉 Store logs in Azure Storage

🔷 Best Practice

- Use Managed Disks
- Avoid ephemeral storage for critical data

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- Use static IP if required

How do you secure secrets using Azure Key Vault in CI/CD pipelines?

In production systems, secret leakage is one of the most common security incidents. The correct design ensures:

- No secrets in code
- No secrets in repo
- No secrets in pipeline variables (plain text)

🔷 Production Architecture Flow

CI/CD pipeline (Azure DevOps / GitHub Actions)
⬇
Authenticates using Managed Identity / Service Principal
⬇
Fetches secrets dynamically from Azure Key Vault
⬇
Injects into runtime (environment variables / app settings)

🔷 Real Implementation Example (Enterprise API)

Scenario:

You have:

- Payment gateway API keys
- Database connection strings
- JWT signing keys

All stored in Azure Key Vault

Pipeline Flow:

1. Pipeline uses Service Principal
2. Key Vault access policy grants GET secret permission

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3. Pipeline step pulls secrets:

```
- task: AzureKeyVault@2
  inputs:
    azureSubscription: 'Prod-Service-Connection'
    KeyVaultName: 'prod-kv'
    SecretsFilter: '*'
```

🔷 Runtime Usage (App Service)

Instead of storing connection string:
Use Key Vault Reference
@Microsoft.KeyVault(SecretUri=https://prod-kv.vault.azure.net/secrets/DbConnection)

No secret ever stored in App Service config.

🔷 Advanced Production Practices

- Enable Key Rotation
- Use Private Endpoint for Key Vault
- Disable public network access
- Enable Audit logs

🔷 Real Failure Scenario

A company stored secrets in:
👉 Azure DevOps variable group (plain text)
One engineer exported logs → secrets leaked.
After fix:
👉 moved to Key Vault + RBAC

7️⃣ Difference between Azure Load Balancer and Application Gateway (real use cases)

This is Layer 4 vs Layer 7 architecture decision

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🔷 Azure Load Balancer (L4)

Operates at:

- TCP
- UDP

Characteristics

- High performance
- No SSL termination
- No URL routing

Real Use Case

👉 Multiplayer gaming backend
👉 TCP-based chat system
👉 VM Scale Set load balancing

🔷 Azure Application Gateway (L7)

Operates at:

- HTTP / HTTPS

Features

- URL-based routing
- SSL termination
- Cookie-based affinity
- Web Application Firewall (WAF)

🔷 Real Production Example

E-commerce platform

Requests:
/api/products → Product service

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/api/orders → Order service
/images → Blob storage

👉 Application Gateway routes based on URL path

🔷 Real Architecture Pattern

Internet
⬇
Application Gateway (WAF)
⬇
Internal Load Balancer
⬇
VM / AKS cluster

🔷 Key Decision

Scenario	Choose
HTTP routing needed	App Gateway
High throughput TCP	Load Balancer
Need WAF	App Gateway
Internal microservices	Internal LB

8️⃣ How do you implement zero-downtime deployment in Azure?

Zero downtime is achieved using traffic switching strategies, not just deployments.

🔷 Strategy 1 — Deployment Slots (App Service)

Flow:

1. Deploy to staging slot
2. Run smoke tests
3. Swap staging → production

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Swap is atomic and instant

🔷 Strategy 2 — Blue/Green Deployment

Two environments:

- Blue (current production)
- Green (new version)

Traffic switch using:

- Front Door
- Traffic Manager

🔷 Strategy 3 — Canary Deployment

Release to:

- 5% users
- Monitor errors
- Increase gradually

Used in:
👉 Payment systems
👉 High-risk deployments

🔷 AKS Rolling Updates

Kubernetes performs:

- Pod-by-pod update
- Keeps min replicas available

🔷 Real Example

Banking system:

- Deploy v2 to staging
- Run automated tests

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- Swap slots
- Monitor Application Insights for 10 mins

If error rate > threshold:
👉 rollback swap

9️⃣ Explain Azure VNet peering vs VPN Gateway

This is network topology design decision.

🔷 VNet Peering

- Connects two VNets inside Azure
- Uses Microsoft backbone
- Very low latency

Real Use Case

Microservices across VNets:
- VNet-A (Payments)
- VNet-B (Orders)

Peering allows internal communication

🔷 VPN Gateway

Used for:
- Azure ↔ On-premises
- Azure ↔ Azure cross-region encrypted

Uses IPSec tunnel over internet

🔷 Real Production Architecture

Enterprise setup:
- On-premises data center

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- Azure VNet Hub

Connected using:
👉 Site-to-Site VPN Gateway

🔷 Performance Comparison

Feature	Peering	VPN
Latency	Low	Medium
Throughput	High	Limited
Encryption	Optional	Encrypted

🔷 Common Mistake

Using VPN between VNets instead of peering → unnecessary cost + latency

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🔟 How do you monitor production issues using Azure Monitor & Log Analytics?

Monitoring is not logs — it is observability system.

🔷 Observability Stack in Azure

Layer	Tool
Infrastructure metrics	Azure Monitor
Logs	Log Analytics
App telemetry	Application Insights
Alerts	Action Groups

🔷 Real Production Monitoring Flow

1. App sends telemetry to Application Insights

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2. Logs stored in Log Analytics Workspace
3. KQL queries detect anomalies
4. Alerts triggered
5. Notification sent to Teams/Slack

🔷 Example KQL Queries

Detect slow API

```
requests
| where duration > 3000
| summarize count() by name
```

Detect failures

```
requests
| where success == false
```

🔷 Alerting Strategy

- Error rate > 2%
- CPU > 80%
- Response time > 2 sec

🔷 Real Incident Example

Production outage:

- API latency increased to 5 seconds
- Alert triggered
- Engineers saw:

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- SQL DTU at 100%
- blocking queries

Fix:
👉 Added index + scaled DB

🔷 Dashboard Design

Production dashboard includes:

- Requests/sec
- Failure rate
- Dependency latency
- DB performance

🔷 Advanced Monitoring

- Distributed tracing
- Correlation ID tracking
- Custom metrics

1️⃣1️⃣ How does Managed Identity work internally?

Managed Identity (MI) eliminates the need to store credentials by using Azure AD–backed identities tied to Azure resources.

🔷 Internal Working (Under the Hood)

When you enable Managed Identity on a resource (VM, App Service, AKS, Function):

1. Azure automatically creates a Service Principal in Azure AD
2. This identity is bound to the resource lifecycle
3. Azure exposes a local metadata endpoint inside the resource:

http://169.254.169.254/metadata/identity/oauth2/token

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4. The application calls this endpoint → gets OAuth2 access token
5. Token is used to call Azure services (Key Vault, Storage, SQL)

🔷 Token Flow (Real Flow)

App Code
⬇
Calls Metadata Endpoint
⬇
Azure AD issues token
⬇
Token used to access resource
⬇
Resource validates token using Azure AD

🔷 Types of Managed Identity

Type	Scope
System Assigned	One identity per resource
User Assigned	Shared across resources

🔷 Real Production Example

AKS microservices accessing Key Vault:

- Each pod uses Pod Identity / Workload Identity
- Gets token via Managed Identity
- Reads secrets from Key Vault

No secrets stored anywhere.

🔷 Security Advantages

- No credentials in code
- Automatic rotation

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- Least privilege via RBAC

🔷 Real Mistake Seen

A team used client secret in config file
Secret expired → production outage
After migration to Managed Identity:
👉 zero secret rotation issues

1️⃣2️⃣ When should you use Azure Functions vs Logic Apps?

This is about code vs workflow orchestration.

🔷 Azure Functions (Code-first compute)

Use when:

- You need custom business logic
- Heavy processing
- Complex transformations
- Integrations using SDKs

Example

Order processing:

- Validate order
- Calculate tax
- Save to DB
- Publish event

👉 Use Azure Function

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🔷 Logic Apps (Workflow orchestration)

Use when:

- Integration workflows
- Event-driven business processes
- No-code / low-code flows

Example

Invoice approval:

- Email received
- Approval request sent
- If approved → send to ERP

👉 Use Logic App

🔷 Key Differences

Factor	Functions	Logic Apps
Coding	Required	Not required
Pricing	Execution-based	Action-based
Dev skill	Developer	Business/IT

🔷 Combined Pattern (Real Enterprise)

Use both together:
Logic App → orchestrates
Function → performs computation

1️⃣3️⃣ How do you design multi-region DR in Azure?

Disaster Recovery is defined by RTO and RPO.

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🔷 DR Strategies

Type	Description
Active-Active	All regions serve traffic
Active-Passive	Primary + standby

🔷 Real Production DR Architecture

For a banking system:

Primary Region

- Central India

Secondary (DR)

- South India

Components

- Azure Front Door (failover routing)
- Azure SQL Failover Group
- Storage GRS replication
- App Service in both regions

🔷 Failover Flow

1. Region failure detected
2. Front Door routes traffic to DR region
3. DB failover group switches primary
4. Application connects using failover endpoint

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🔷 Data Consistency Consideration

Mode	Trade-off
Synchronous	zero data loss but slower
Asynchronous	small data loss but faster

🔷 Real Incident Example

A logistics company had:

- Single region DB
- No geo-replication

Region outage → 6 hours downtime
After redesign:
👉 DR region + failover group → <10 min downtime

1️⃣4️⃣ What are Availability Sets vs Availability Zones (real impact)?

These are fault isolation strategies.

🔷 Availability Set

Protects against:

- Rack failure
- Power unit failure

Uses:

- Fault domains
- Update domains

Example

3 VMs placed in different racks inside same data center

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🔷 Availability Zones

Physically separate data centers within region
Each zone has:

- Independent power
- Cooling
- networking

🔷 Real Impact Comparison

Scenario	Availability Set	Availability Zone
Rack failure	safe	safe
Data center failure	down	✅ safe
Region failure		

🔷 Production Example

Payment system:

- 3 AKS node pools across zones
- Zone redundancy ensures app works even if one DC goes down

🔷 Recommendation

Modern production systems:
👉 Always prefer Availability Zones
Availability Sets are legacy approach

1️⃣5️⃣ How do you secure APIs using Azure API Management?

API Management (APIM) is the API gateway + security layer.

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🔷 Security Layers in APIM

1. Authentication

- OAuth2 / OpenID Connect
- Azure AD integration
- JWT validation policy

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2. Authorization

- Role-based access
- Subscription keys

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3. Traffic Protection

- Rate limiting
- Quotas
- Spike protection

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4. Network Security

- Private endpoint
- IP filtering
- VNet integration

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5. Web Application Firewall

Integrate with Application Gateway WAF

🔷 Real Production Example

A fintech exposes APIs to partners:
Flow:

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Client → APIM → Backend microservices
APIM does:

- Validate JWT token
- Check subscription
- Rate limit (1000 req/min)
- Log request

🔷 Example Policy

```
<validate-jwt header-name="Authorization">
  <openid-config url="https://login.microsoftonline.com/..."/>
</validate-jwt>
```

🔷 Advanced Enterprise Features

- API versioning
- Transformation (XML → JSON)
- Mock responses for testing
- Developer portal

🔷 Real Security Incident

Without APIM:

- APIs exposed directly
- No rate limit
- Bot traffic caused outage

After APIM:
👉 rate limiting + WAF blocked malicious traffic

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1️⃣6️⃣ Explain RBAC vs Azure Policy with real enterprise examples

This is a critical governance topic in enterprise Azure.

🔷 Core Difference

Feature	RBAC	Azure Policy
Controls	Who can do	What can be done
Level	Access control	Resource compliance
Scope	User/Group/Service	Resource configuration

🔷 RBAC (Role-Based Access Control)

Controls permissions.

Example Roles:

- Reader
- Contributor
- Owner
- Custom roles

Real Enterprise Example

A fintech company:

- Developers → Contributor on Dev resource group
- QA → Reader
- Production → Only DevOps team has access

👉 This is RBAC

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🔷 Azure Policy

Controls rules on resources

Examples:

- Only allow resources in India regions
- Enforce tagging
- Block public IP creation
- Force HTTPS

Real Enterprise Example

Bank enforces:

- No public IPs
- Storage must be encrypted
- Only Standard SKU allowed

👉 Done using Azure Policy

🔷 Combined Usage (Real Pattern)

RBAC + Policy together:

- RBAC controls who can deploy
- Policy ensures what they deploy is compliant

1️⃣7️⃣ How do you handle data encryption at rest and in transit in Azure?

Security requires defense-in-depth encryption strategy.

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🔷 Encryption at Rest

Storage

- Azure Storage Service Encryption (SSE)
- Uses Microsoft-managed keys by default

Advanced

- Customer Managed Keys (CMK) in Key Vault

Databases

- Azure SQL → Transparent Data Encryption (TDE)
- Cosmos DB → always encrypted

🔷 Encryption in Transit

All communication uses:

- HTTPS (TLS 1.2+)
- VPN/IPSec tunnels
- Private endpoints

🔷 Real Production Architecture

A healthcare system:

- App Service → HTTPS only
- Database → TDE enabled
- Storage → CMK encryption
- Internal traffic → Private endpoints

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🔷 Additional Security

- Always Encrypted for sensitive columns
- Double encryption for compliance workloads

1️⃣8️⃣ What challenges did you face while migrating to Azure?

Migration is never just lift-and-shift.

🔷 Real Challenges Seen in Production

1. Legacy Monolith Systems

- Tight coupling
- Hardcoded IPs
- No scaling

👉 Solution:
Refactor to microservices gradually

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2. Network Complexity

- On-prem firewall rules
- Latency issues

👉 Solution:
ExpressRoute / VPN + hub-spoke network

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3. Cost Explosion

Lift-and-shift VMs → high cost
👉 Solution:
Move to PaaS (App Service, Azure SQL)

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4. Security Redesign

On-prem identity ≠ cloud identity
👉 Solution:
Azure AD, Managed Identity

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5. Data Migration Downtime

Large DB (2TB+)
👉 Solution:

- Online replication
- Cutover window

🔷 Real Case Study

Retail company migration:
Initial:

- 60 VMs

After modernization:

- 12 App Services
- 2 Azure SQL
- 1 Redis

👉 Cost reduced 50%
👉 Performance improved

1️⃣9️⃣ Azure DevOps CI/CD vs GitHub Actions

Both are enterprise-grade, but used differently.

🔷 Azure DevOps

Full ALM platform:

- Boards (work tracking)
- Repos
- Pipelines

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- Test Plans

Used by large enterprises

🔷 GitHub Actions

- GitHub-native CI/CD
- YAML-based workflows
- Marketplace integrations

Used by modern cloud-native teams

🔷 Pipeline Comparison

Feature	Azure DevOps	GitHub Actions
UI pipelines	Yes	No
YAML pipelines	Yes	Yes
Repo integration	Azure Repos	GitHub
Enterprise ALM	Strong	Moderate

🔷 Real Use Case

- Enterprise bank → Azure DevOps
- Startup SaaS → GitHub Actions

🔷 Hybrid Pattern

Many companies use:

- GitHub (code)
- Azure DevOps (release pipelines)

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2️⃣0️⃣ How do you troubleshoot slow Azure SQL performance?

This is a very common production issue.

🔷 Step-by-Step Production Debugging

Step 1: Check resource usage

- DTU/CPU %
- Memory pressure
- IO

Step 2: Query Performance Insight

Identify:

- Slow queries
- High CPU queries
- Missing indexes

Step 3: Blocking & Deadlocks

Use query:
SELECT * FROM sys.dm_tran_locks

Step 4: Index Optimization

- Add missing indexes
- Remove unused indexes

Step 5: Scale Database

- Increase DTU/vCore

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- Use elastic pool

🔷 Real Incident Example

E-commerce site slow checkout:
Issue:

- Missing index on Orders table

Fix:

- Added composite index

Result:

- Query reduced from 4 sec → 200 ms

🔷 Advanced Optimization

- Query store
- Partitioning
- Read replicas
- Caching (Redis)

2️⃣1️⃣ Explain Azure Storage redundancy options (LRS, ZRS, GRS)

Storage redundancy is about durability vs availability vs cost.

🔷 Types of Redundancy

Type	Copies	Location	Use Case
LRS	3	Same datacenter	Dev/Test, low cost
ZRS	3	Across availability zones	Production apps
GRS	6	Primary + secondary region	Disaster recovery

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GZRS	6	Zones + geo region	Mission critical

🔷 Real Production Decision

Scenario: E-commerce product images

- Must be highly available
- But can tolerate small delay during disaster

👉 Use ZRS

Scenario: Financial transaction logs

- Must never lose data
- Must survive region failure

👉 Use GRS or GZRS

🔷 Trade-offs

Factor	LRS	ZRS	GRS
Cost	Low	Medium	High
Region failure safe			✅
Zone failure safe		✅	✅

2️⃣2️⃣ How do you design event-driven architecture in Azure?

Event-driven systems decouple services and improve scalability.

🔷 Azure Event Stack

Service	Purpose
Event Grid	Event routing

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Service Bus	Reliable messaging
Event Hub	Streaming

🔷 Real Architecture Example

E-commerce Order Flow
User places order → event published

Flow

1. Order Service → publishes event to Event Grid
2. Multiple subscribers:
    - Billing Service
    - Inventory Service
    - Email Service
3. Each processes independently

🔷 Benefits

- Loose coupling
- Async processing
- Fault isolation

🔷 Real Failure Handling

If email service fails:
👉 message stays in queue
👉 retry later

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2️⃣3️⃣ How do you manage infrastructure as code in Azure?

IaC ensures repeatable, version-controlled infrastructure.

🔷 Tools

- ARM Templates
- Bicep (recommended)
- Terraform (multi-cloud)

🔷 Real Enterprise Setup

Repo structure:

```
infra/
  dev/
  qa/
  prod/
modules/
  network/
  appservice/
  sql/
```

🔷 CI/CD Flow

1. Developer commits infra code
2. Pipeline validates (lint + security scan)
3. Deploys using:
    - az deployment group create

🔷 Benefits

- No manual portal work

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- Repeatable environments
- Easy rollback

🔷 Real Incident Avoided

Manual config mistake caused:
👉 open database to internet
After IaC:
👉 policy + templates prevented misconfig

2️⃣4️⃣ What is Azure Blue-Green vs Canary deployment?

Deployment strategies to reduce production risk.

🔷 Blue-Green Deployment

Two environments:

- Blue → current production
- Green → new version

Switch traffic instantly

Real Use Case

Banking system:

- Deploy v2 in Green
- Validate
- Switch traffic

🔷 Canary Deployment

Gradual rollout

- 5% users

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- 20%
- 50%
- 100%

Real Use Case

Payment gateway:

- New fraud detection logic
- Release to 5% traffic
- Monitor fraud rate

🔷 Comparison

Factor	Blue-Green	Canary
Risk	Medium	Low
Complexity	Low	High
Rollback	Instant	Gradual

2️⃣5️⃣ How do you secure workloads in AKS?

AKS security is multi-layered.

🔷 Key Security Layers

1. Cluster Security

- RBAC enabled
- Azure AD integration

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2. Network Security

- Network Policies
- Private cluster

3. Identity

- Workload Identity (Managed Identity for pods)

4. Secrets

- Azure Key Vault + CSI driver

5. Image Security

- Use private container registry (ACR)
- Image scanning

🔷 Real Production Setup

Banking microservices in AKS:

- Private cluster
- No public API server
- All ingress via Application Gateway
- Secrets from Key Vault

🔷 Common Mistakes

Exposing AKS publicly
Storing secrets in config map
No network segmentation

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2️⃣6️⃣ How do you control resource sprawl in large Azure subscriptions?

At scale, uncontrolled resources = cost + security risk

🔷 Governance Model

1. Naming convention

app-env-region-type

Example:
orders-prod-centralindia-appsvc

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2. Tagging

Mandatory tags:

- Owner
- CostCenter
- Environment
- Project

---

3. Azure Policies

- Block public IP
- Enforce region
- Enforce SKU

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4. Budgets + Alerts

Notify when cost crosses threshold

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5. Resource Locks

Prevent accidental deletion

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🔷 Real Example

Enterprise had:
👉 4000+ resources
After governance:
👉 reduced 22% unused resources

2️⃣7️⃣ Explain Azure Private Endpoint vs Service Endpoint

Both provide secure access to PaaS services, but architecture differs.

🔷 Service Endpoint

- Extends VNet identity to Azure service
- Traffic still goes via public endpoint

🔷 Private Endpoint

- Assigns private IP from VNet
- Fully private network access

🔷 Real Comparison

Feature	Service Endpoint	Private Endpoint
IP Type	Public	Private
Security	Medium	High
Use case	Basic secure access	Sensitive workloads

🔷 Real Production Example

Healthcare system:

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- SQL DB via Private Endpoint
- Storage via Private Endpoint

No public exposure

2️⃣8️⃣ How do you implement centralized logging in Azure?

Centralized logging is critical for observability + compliance.

🔷 Architecture

All services send logs to:
👉 Log Analytics Workspace

🔷 Sources

- App Service logs
- AKS logs
- VM logs
- API Management logs
- Azure Firewall logs

🔷 Flow

Resource → Diagnostic Settings → Log Analytics

🔷 Real Dashboard

- Error rate
- Request latency
- DB failures

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- Security logs

🔷 Alerting

KQL query triggers alert:
exceptions
| summarize count() by type

2️⃣9️⃣ How do you estimate Azure pricing before production?

Never go to production without cost modeling.

🔷 Tools Used

- Azure Pricing Calculator
- TCO Calculator
- Load testing

🔷 Estimation Steps

1. Estimate traffic (RPS)
2. Estimate compute size
3. Storage size
4. Data transfer
5. Backup cost

🔷 Real Example

SaaS CRM expected:

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- 10k users
- 100 RPS

Cost estimated:

- App Service Plan
- Azure SQL
- Redis
- Storage

Final estimate:
👉 ₹8 lakhs/month

🔷 Best Practice

Always add 20–30% buffer

3️⃣0️⃣ What Azure design decision did you regret — and why?

This is a behavioral + architectural maturity question

🔷 Strong Real Answer Example

“In one project, we deployed a monolithic .NET application on multiple Azure VMs behind a Load Balancer.
At the time it seemed simple, but over time we faced:

- Difficult scaling
- Complex deployments
- High maintenance cost
- OS patching overhead

Later, we re-architected to:

- Azure App Service (PaaS)
- Azure SQL

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- Redis cache

This reduced:

- Infrastructure overhead by 60%
- Deployment time from 30 mins to 5 mins
- Improved scalability and reliability

The key lesson:
👉 Always prefer PaaS unless you have a strong reason for IaaS."

For Complete Interview Question Answer: https://topmate.io/sandeeppal

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